CLIMATE JAM 2026 · SPRINT 1 · IDEATION SUBMISSION

FarmBotVille: Resilencity Blackfox Studios

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WORKING PROJECT TITLE & TEAM NAME

Project title: FarmBotVille: Resilencity (Innovation from Necessity)

Team name: Blackfox Studios

Team: Andrew Powers, design, systems, narrative, and direction. Claude, engineering seat (game logic, HTML5, itch.io build). Makko.ai, art department (style locked sprite collection).

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OVERVIEW

Q: What is the game idea? A short summary. Who is the player and what do they do? What makes it engaging? What is the genre and platform?

A: FarmBotVille: Resilencity is a narrative survival and engineering sim. You play Tyler King, a fifteen year old who turns the dump next to his house into a supply depot and his backyard into a system that feeds his family.

The player scavenges waste and runs it through a chain of machines to close one loop: bottle to filament to part to crop to food to compost to soil. What makes it engaging is watching trash become dinner and understanding exactly how it happened, all under a countdown that keeps taking resources away.

Genre: systems and resource simulation with a maker core. Platform: browser based (HTML5), submitted on itch.io.

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GAMEPLAY

Q: How is the game played? Core loop, player goal, win and lose conditions, key mechanics, player actions, obstacles.

A: Core concept and gameplay loop. The heart of the game is the Bottle Loop. Every machine follows one rule: inputs plus a skill plus energy produce outputs over time. The machines chain together into a closing cycle. Polyformer turns bottles into filament, the printer turns filament into a part, the FarmBot turns parts and water and seed into a crop, the crop feeds the family, and the composter returns the waste to the soil.

Player goal, win and lose conditions. The goal is self reliance, not wealth. You win when the loop closes and the household survives the crisis on what the backyard produces. You lose after two hungry nights, when the pantry empties before the loop is running.

Key mechanics and systems. A ticking clock advances time, so plants grow, the battery drains, compost heats, and the rain barrel fills whether or not you act. Progression works by Compound Learning, where skills combine to unlock recipes. Guidance comes from Stone, an offline AI companion who reads the system and offers the Tri Path choice.

Player actions and interactions. Scavenge the dump, feed materials into machines, allocate scarce battery and water between competing systems, and make Tri Path decisions: the scrap yard hack (fast, cheap, may break), the ancestral method (hand water daily, costs your time), or the code compliant build (slow, reliable).

Obstacles and challenges. The Crisis Timeline strips a resource on schedule: inflation, then the SNAP cut, then delivery isolation, all against a six day pantry clock. Every stage removes an outside crutch and forces deeper reliance on the loop.

Status note. This is not aspirational text. It is a running build, verified across 1,000 simulations: optimal play wins 98.8 percent of the time, sloppy play wins 83.8 percent.

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CLIMATE THEME & IMPACT

Q: How does the game connect to Climate Jam themes? What topic are you exploring? What is your real world basis? What is your message?

A: Climate topic. The circular economy at residential scale: waste diversion, food security, and the environmental justice reality of a neighborhood designed to consume rather than produce.

Real world basis and inspiration. The basis is literal. The game is the playable feasibility study of Re+Filament Farms, an actual circular economy house being developed in Des Moines, Washington. The machines are real: the FarmBot is an open source farming robot, the Polyformer turns real bottles into real filament, and Stone stands in for a real Raspberry Pi.

Intended message and call to action. The reframe is the message: waste is not waste, it is feedstock, and a neighborhood that looks like a trap can be read as a supply of everything a resilient home needs. When the systems fall, that is when we build. Every backyard can become a small utility.

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ART & TONE

Q: What is the look, feel, and tone? Visual style, sound, narrative, mood, and reference games or images.

A: Visual style. Solarpunk crossed with Afrofuturism and scrap yard science. Warm, hopeful, and resourceful rather than grim, even though the premise is a collapse. Icon first and legible, so a partner who does not play games can watch the loop and understand it at a glance.

Music and sound design. Kept minimal for the jam and oriented toward clear feedback: a machine completing, a resource arriving, a crisis stage triggering.

Narrative and dialog. Delivered mainly through Stone, whose voice blends a patient science mentor with a guide, and through short exchanges that keep the human stakes (the grandparents, the block) present.

Mood and emotional experience. Competence under pressure. The specific satisfaction of making something work with what is on hand.

Reference points. Stardew Valley for the warmth of a home base, Dr. Stone for the joy of chained invention, Shapez.io for a system laid bare, Terra Nil for the restoration ethic. Current build uses simple vector shapes at fixed anchor points; Makko sprites replace them at those same anchors.

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INNOVATION & DESIGN APPROACH

Q: What makes the game stand out? Unique mechanics, experimental choices, what feels new.

A: The signature choice is architectural. The game's state variables are named as real MQTT topics, for example resilencity/tyler/battery/soc, so the game is a digital twin in waiting. The same build can later drive or be driven by the real house with no rewrite. A Raw Topics toggle in the build shows this live on screen.

Two more choices set it apart. Compound Learning makes the tech tree, the story, and the curriculum a single structure, so learning is the mechanic rather than a layer on top. Tri

Path builds cultural range into the core loop, offering scrap yard, ancestral, and code compliant solutions side by side as mechanics with real tradeoffs.

The combination is the new thing: a resource conversion sim, wrapped in a restoration goal, inside a crisis narrative, grounded in a real place and real hardware.

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TEAM & COLLABORATION

Q: Describe the team. Roles, communication tools, collaboration methods, version control.

A: Primary roles. Andrew Powers leads design, systems, narrative, and direction. Claude fills the engineering seat, writing all logic, the HTML5, and the itch.io build. Makko.ai is the art department, with a possible local collaborator through its founder.

Communication and workflow. The single source of truth is the onboarding and definitions document plus the system architecture diagram, so any collaborator can join with no prior context. Planning and documents live in Google Workspace.

Collaboration methods and version control. The itch.io project page is the public home of the build. The engine lives in one commented engine.js file, kept separate from the interface, which also serves as future curriculum material. Given the small team, the model is lightweight and asynchronous.

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PRODUCTION PLAN & MILESTONES

Q: How will you complete the project? Milestones, task breakdown, known risks, and scope control.

A: Resolved first gate. The original open question, whether to build in Makko or Scratch, is settled. Claude writes HTML5 directly, so Scratch is dropped and a playable greybox already runs.

Remaining milestones.

1. Playtest and tune the balance. All knobs live in one TUNE block, so notes become one line changes.
2. Build the Makko art Collection: Tyler, Stone, the machines, the backyard, the dump.
3. Swap the sprites over the greybox shapes at the fixed anchor points.
4. Polish for legibility and add feedback sound.
5. Submit early, ahead of the deadline.

Scope control, what gets cut and in what order. Multiplayer, then save and load, then the Ada and Kim characters, then a live AI Stone (the scripted Stone stays), then the full dual cycle, then the Kinetic Gym. What is never cut is the Bottle Loop closing once inside a single crisis beat. That is the whole point, and everything else is negotiable around it.

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Blackfox Studios · Andrew Powers · andrew@blackfoxstudios.org A playable feasibility study of house scale circularity.